

The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP5010				
Subject Title	Artificial Intelligence for Materials Science				
Credit Value	3				
Level	5				
Pre-requisite / Co-requisite/ Exclusion	Nil				
Objectives	To introduce the fundamentals of artificial intelligence algorithms and their applications in materials science.				
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Understand the artificial intelligence (AI) algorithms, concepts, and workflow. (b) extract, visualize and process data from online databases dedicated to materials scientists and microelectronics engineers; (c) select AI models to solve specific problems; (d) implement various machine learning models in Python; and (e) evaluate and improve the performance of a model.				
Subject Synopsis/ Indicative Syllabus	Linear regression and classification: databases, feature, representation, training set, error functions, gradient descent, univariate and multivariate regression, logistic regression Neural networks: activation functions, backpropagation, feed-forward networks, recurrent networks, deep learning Unsupervised learning: clustering, dimension reduction Theory of machine learning: overfitting, validation, regularization, generalization, bias-variance Application in physical science: Materials modeling (DFT+AI); Materials dataset and data mining; Materials project and its applications (2D materials and electro-catalysts); Bulk modulus, bandgap, energy, force, molecular dynamics				
Teaching/Learning Methodology	The course will make use of Anaconda as well as its features for collaboration and publishing. Students will learn how to program in Python with Jupyter notebook and use external libraries such as Numpy, Scipy, Scikit-learn, Pandas, and Tensorflow. Lecture: The concepts will be presented in class including some illustrative examples and algorithms related to solving problems in physics or engineering. Computer laboratory: During laboratory sessions, students will write small programs individually or in groups to gain a deeper understanding of the topics discussed during the lectures. These sessions aim at developing the students' intuition and critical thinking upon discussion and exchange of ideas to engage in the proposed activities.				
Assessment Methods in Alignment with Intended Learning	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
			a	b	c

Outcomes	1. Continuous Assessments	50%	✓	✓	✓	✓	✓
	2. Project	50%	✓	✓	✓	✓	✓
	Total	100 %					
	Continuous assessments include training of computer programming in the computer laboratory sessions that are used to check and grade students’ progress throughout the semester. The project will be assigned during the second half of the semester. Students will use their knowledge gained in the class/lab to design a machine learning program to tackle a problem related to physical science.						
Student Study Effort Expected	Class contact:						
	▪ Lecture					26 Hrs.	
	▪ Computer Laboratory					13 Hrs	
	Other student study efforts:						
	▪ Reading and self-study					81 Hrs.	
	The total student study effort					120 Hrs.	
Reading List and References	Simon Rogers, and Mark Girolami, A first course in machine learning, Chapman and Hall/CRC, 2nd Ed., 2016. A. Géron, Hands-on machine learning with Scikit-Learn and TensorFlow: concepts, tools, and techniques to build intelligent systems, 1st Ed., O'Reilly Media, 2017. Yuan Cheng, Tian Wang and Gang Zhang, Artificial Intelligence for Materials Science, Springer, 2021. Andrew Ng, Machine Learning, MOOC from Stanford (CS229). Yaser S. Abu-Mostafa, Learning From Data, MOOC from Caltech.						